```
import numpy as np
import pandas as pd
import re
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

import nltk
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
print(stopwords.words('english'))
```

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', '
```

Data Preprocessing

```
true_df = pd.read_csv('/content/True.csv')
fake_df = pd.read_csv('/content/Fake.csv')
```

```
print(true_df.head())
print(fake_df.head())
```

```
                                               title  ...                date
0  As U.S. budget fight looms, Republicans flip t...  ...  December 31, 2017
1  U.S. military to accept transgender recruits o...  ...  December 29, 2017
2  Senior U.S. Republican senator: 'Let Mr. Muell...  ...  December 31, 2017
3  FBI Russia probe helped by Australian diplomat...  ...  December 30, 2017
4  Trump wants Postal Service to charge 'much mor...  ...  December 29, 2017

[5 rows x 4 columns]
                                               title  ...               date
0   Donald Trump Sends Out Embarrassing New Year’...  ...  December 31, 2017
1   Drunk Bragging Trump Staffer Started Russian ...  ...  December 31, 2017
2   Sheriff David Clarke Becomes An Internet Joke...  ...  December 30, 2017
3   Trump Is So Obsessed He Even Has Obama’s Name...  ...  December 29, 2017
4   Pope Francis Just Called Out Donald Trump Dur...  ...  December 25, 2017

[5 rows x 4 columns]
```

```
true_df['label'] = 0
fake_df['label'] = 1
```

1. True News = 0
2. Fake News = 1

```
print(true_df.head())
print(fake_df.head())
```

```
                                               title  ... label
0  As U.S. budget fight looms, Republicans flip t...  ...     0
1  U.S. military to accept transgender recruits o...  ...     0
2  Senior U.S. Republican senator: 'Let Mr. Muell...  ...     0
3  FBI Russia probe helped by Australian diplomat...  ...     0
4  Trump wants Postal Service to charge 'much mor...  ...     0

[5 rows x 5 columns]
                                               title  ... label
0   Donald Trump Sends Out Embarrassing New Year’...  ...     1
1   Drunk Bragging Trump Staffer Started Russian ...  ...     1
2   Sheriff David Clarke Becomes An Internet Joke...  ...     1
3   Trump Is So Obsessed He Even Has Obama’s Name...  ...     1
4   Pope Francis Just Called Out Donald Trump Dur...  ...     1

[5 rows x 5 columns]
```

```
true_df.isnull().sum()
```

	0
title	0
text	0
subject	0
date	0
label	0

dtype: int64

```
fake_df.isnull().sum()
```

	0
title	0
text	0
subject	0
date	0
label	0

dtype: int64

```
print(true_df.shape)
print(fake_df.shape)
```

```
(21417, 5)
(23481, 5)
```

```
print(true_df['date'])
```

```
0        December 31, 2017
1        December 29, 2017
2        December 31, 2017
3        December 30, 2017
4        December 29, 2017
                ...
21412      August 22, 2017
21413      August 22, 2017
21414      August 22, 2017
21415      August 22, 2017
21416      August 22, 2017
Name: date, Length: 21417, dtype: object
```

```
print(true_df.columns)
print(fake_df.columns)
```

```
Index(['title', 'text', 'subject', 'date', 'label'], dtype='object')
Index(['title', 'text', 'subject', 'date', 'label'], dtype='object')
```

```
news_dataset = pd.concat([true_df, fake_df], axis=0)
```

```
news_dataset = news_dataset.sample(frac = 1, random_state=42).reset_index(drop=
```

```
news_dataset["id"] = news_dataset.index
```

```
news_dataset.head()
```

	title	text	subject	date	label	id
0	BREAKING: GOP Chairman Grassley Has Had Enoug...	Donald Trump s White House is in chaos, and th...	News	July 21, 2017	1	0
1	Failed GOP Candidates Remembered In Hilarious...	Now that Donald Trump is the presumptive GOP n...	News	May 7, 2016	1	1
2	Mike Pence's New DC Neighbors Are HILARIOUSLY...	Mike Pence is a huge homophobe. He supports ex...	News	December 3, 2016	1	2
	California AG pledges	SAN FRANCISCO (Reuters) -		October 6,		

Next steps: Generate code with news_dataset New interactive sheet

```
cols = ["id"] + [col for col in news_dataset.columns if col != "id"]
news_dataset = news_dataset[cols]
```

```
news_dataset.head()
```

	id	title	text	subject	date	label
0	0	BREAKING: GOP Chairman Grassley Has Had Enoug...	Donald Trump s White House is in chaos, and th...	News	July 21, 2017	1
1	1	Failed GOP Candidates Remembered In Hilarious...	Now that Donald Trump is the presumptive GOP n...	News	May 7, 2016	1
2	2	Mike Pence's New DC Neighbors Are HILARIOUSLY...	Mike Pence is a huge homophobe. He supports ex...	News	December 3, 2016	1
		California AG pledges	SAN FRANCISCO (Reuters) -		October 6,	

Next steps: Generate code with news_dataset New interactive sheet

```
news_dataset.to_csv('F:\Machine Learning Projects\Project 4 - Fake News Predict
```

```
<>:1: SyntaxWarning: invalid escape sequence '\M'
<>:1: SyntaxWarning: invalid escape sequence '\M'
/tmp/ipykernel_30045/4277763696.py:1: SyntaxWarning: invalid escape sequence '\M
  news_dataset.to_csv('F:\Machine Learning Projects\Project 4 - Fake News Predic
```

merging the news title and news text

```
news_dataset['content'] = news_dataset['title'] + ' ' + news_dataset['text']
```

```
news_dataset.head()
```

	id	title	text	subject	date	label	content
0	0	BREAKING: GOP Chairman Grassley Has Had Enoug...	Donald Trump s White House is in chaos, and th...	News	July 21, 2017	1	BREAKING: GOP Chairman Grassley Has Had Enoug...
1	1	Failed GOP Candidates Remembered In Hilarious...	Now that Donald Trump is the presumptive GOP n...	News	May 7, 2016	1	Failed GOP Candidates Remembered In Hilarious...
		Mike Pence's	Mike Pence is a huge				Mike Pence's

Next steps: Generate code with news_dataset New interactive sheet

```
# Separating the data and label
X = news_dataset.drop(columns='label', axis=1)
Y = news_dataset['label']
```

```
print(X)
print(Y)
```

```
          id  ...                                            content
0          0  ...   BREAKING: GOP Chairman Grassley Has Had Enoug...
1          1  ...   Failed GOP Candidates Remembered In Hilarious...
2          2  ...   Mike Pence’s New DC Neighbors Are HILARIOUSLY...
3          3  ...  California AG pledges to defend birth control ...
4          4  ...  AZ RANCHERS Living On US-Mexico Border Destroy...
...      ...  ...                                                ...
44893  44893  ...  Nigeria says U.S. agrees delayed $593 million ...
44894  44894  ...  Boiler Room #62 – Fatal Illusions Tune in to t...
44895  44895  ...  ATHEISTS SUE GOVERNOR OF TEXAS Over Display on...
44896  44896  ...  Republican tax plan would deal financial hit t...
44897  44897  ...  U.N. refugee commissioner says Australia must ...

[44898 rows x 6 columns]
0        1
1        1
2        1
3        0
```

```
4        1
        ..
44893    0
44894    1
44895    1
44896    0
44897    0
Name: label, Length: 44898, dtype: int64
```

Stemming:

Stemming is the process of reducing a word to its Root word

```
port_stem = PorterStemmer()
```

```
def stemming(content):
    stemmed_content = re.sub('[^a-zA-Z]',' ', content)
    stemmed_content = stemmed_content.lower()
    stemmed_content = stemmed_content.split()
    stemmed_content = [port_stem.stem(word) for word in stemmed_content if not
    stemmed_content = ' '.join(stemmed_content)
    return stemmed_content
#
```

```
news_dataset['content'] = news_dataset['content'].apply(stemming)
```

```
print(news_dataset['content'])
```

```
0        break gop chairman grassley enough demand trum...
1        fail gop candid rememb hilari mock eulog video...
2        mike penc new dc neighbor hilari troll homopho...
3        california ag pledg defend birth control insur...
4        az rancher live us mexico border destroy nanci...
                              ...
44893    nigeria say u agre delay million fighter plane...
44894    boiler room fatal illus tune altern current ra...
44895    atheist sue governor texa display capitol grou...
44896    republican tax plan would deal financi hit u u...
44897    u n refuge commission say australia must stop ...
Name: content, Length: 44898, dtype: object
```

```
X = news_dataset['content'].values
Y = news_dataset['label'].values
```

```
print(X)
```

```
['break gop chairman grassley enough demand trump jr testimoni donald trump whit
 'fail gop candid rememb hilari mock eulog video donald trump presumpt gop nomin
 'mike penc new dc neighbor hilari troll homophob bigot mike penc huge homophob
```

```
 ...
 'atheist sue governor texa display capitol ground convinc freedom religion grou
 'republican tax plan would deal financi hit u univers washington reuter republi
 'u n refuge commission say australia must stop unfold humanitarian crisi png sy
```

```
print(Y)
```

```
[1 1 1 ... 1 0 0]
```

```
Y.shape
```

```
(44898,)
```

```
# Converting the textual data to numerical data

vectorizer = TfidfVectorizer()
vectorizer.fit(X)

X = vectorizer.transform(X)
```

```
print(X)
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
        with 6904081 stored elements and shape (44898, 89868)>
  Coords        Values
  (0, 570)      0.06055486453372135
  (0, 2286)     0.04253552689623891
  (0, 2301)     0.025666202113816185
  (0, 3031)     0.063093661763644
  (0, 3388)     0.047259753130124114
  (0, 6574)     0.07041583382179394
  (0, 7853)     0.0898677958851911
  (0, 7886)     0.059913473329118644
  (0, 9454)     0.048097596645320734
  (0, 10324)    0.08231690448262517
  (0, 11174)    0.03296114815015033
  (0, 11631)    0.04919300578107092
  (0, 12522)    0.05138750652591612
  (0, 12642)    0.06680808165549404
  (0, 13496)    0.06811316396970238
  (0, 13713)    0.07112856852592032
  (0, 13991)    0.053000278097355684
  (0, 14098)    0.037972261733148126
  (0, 14836)    0.0869210456407381
  (0, 14928)    0.06842646468851935
  (0, 15149)    0.04165264736324387
  (0, 15153)    0.051876038877662234
  (0, 15333)    0.08494945735141231
  (0, 15663)    0.056535937666043194
  (0, 15765)    0.029199968119368196
  :     :
```

```
  (44897, 77773)        0.03757368913834899
  (44897, 77960)        0.01293068706666485
  (44897, 78688)        0.042561076129468856
  (44897, 78704)        0.040544018114292965
  (44897, 79058)        0.05779728742472851
  (44897, 80447)        0.04313047656027923
  (44897, 81481)        0.07874060468713938
  (44897, 81621)        0.04451315198091657
  (44897, 82255)        0.03796982127137953
  (44897, 82341)        0.030435493815230488
  (44897, 82450)        0.03850229932669336
  (44897, 83260)        0.033882067136460456
  (44897, 83611)        0.02756733398300722
  (44897, 84581)        0.020184813480884824
  (44897, 84769)        0.13810212379455322
  (44897, 85054)        0.018516634241257903
  (44897, 85198)        0.01690775168103281
  (44897, 86130)        0.030132938735777468
  (44897, 86169)        0.040827065890862685
  (44897, 86180)        0.059445897055736864
  (44897, 86481)        0.04853978108940374
  (44897, 86551)        0.015240029412780992
  (44897, 86624)        0.029560471446670476
  (44897, 87936)        0.024069000877548073
  (44897, 88056)        0.021027619607628736
```

Splitting the dataset to training & test data

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, strati
```

Applying Logistic Regression Model

```
model = LogisticRegression()
```

```
model.fit(X_train, Y_train)
```

```
▾ LogisticRegression
LogisticRegression()
```

Evaluating accuracy score

```
X_train_prediction = model.predict(X_train)
training_data_accuracy = accuracy_score(X_train_prediction, Y_train)
```

```
print('Accuracy score of the training data : ', training_data_accuracy)
```

```
Accuracy score of the training data :  0.9917033242385433
```

```
X_test_prediction = model.predict(X_test)
test_data_accuracy = accuracy_score(X_test_prediction, Y_test)
```

```
print('Accuracy score of the test data : ', test_data_accuracy)
```

```
Accuracy score of the test data :  0.9876391982182628
```

Building a Predicition system

```
X_new = X_test[4]

prediction = model.predict(X_new)
print(prediction)

if (prediction[0]==0):
  print('The news is Real')
else:
  print('The news is Fake')
```

```
[0]
The news is Real
```

```
print(Y_test[4])
```

```
0
```

Start coding or generate with AI.